RAJALAKSHMI ENGINEERING COLLEGE

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H2

Started on	Thursday, 6 August 2026, 10:27 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:04 AM
Time taken	36 mins 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
int main() {
    int num;
    int count=0;
    scanf("%d",&num);
    for (int i=1;i<=num;++i)
    {
        count++;
        if (num%i==0)
        {
            count++;
        }
    }
    count+=num;
    count++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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